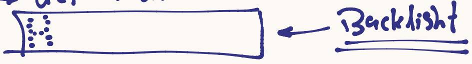
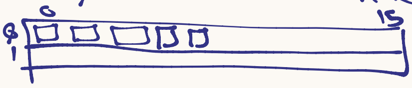
- Display H.W
  - ① led → Notification
  - ② segment → Number
  - ③ screen → like → LCD

* LCD: liquid crystal Display
  - ① LCD → default on → word → off

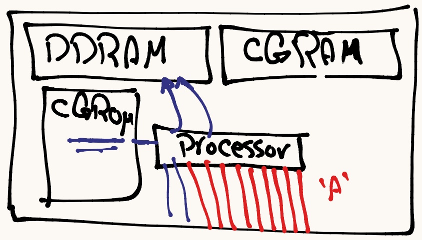


  - ② LCD → type: 16*2 | 20*4
    - 16 → No of digit
    - 2 → No of line

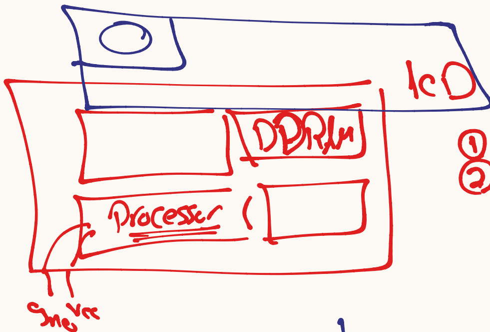


  - ③ Mc & LCD → Parrallel Communication

[2] LCD Digram overview



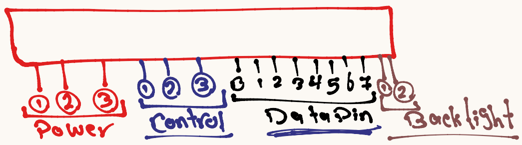
① DDRAM: Data Display Ram
* volatile * Connect with screen Par



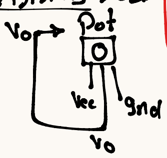
① Power
② clock

LCD
- ① Pinout
- ② Digram over view
- ③ How to work

* Pinout



Power Pin
- ① → gnd ≃ Vss
- ② → Vcc ≃ VEE
- ③ → Vo → Vo
  - Contrast →
  - ① Display → off

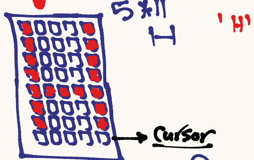


* Backlight
* ① BkA → Vcc
* ② BkC → gnd

Control Pin
- ① Register select
  - RS → Display Data (1)
  - → Instruction Command (0)
- ② Read/write
  - R/W̄: 0 → write
  - : 1 → Read.
  - R/W̄ → gnd
- ③ Enable
  - E = 1 → wait → E = 0

* Data Pin
- ① you send
  - → Display data
  - → Command
  - on Pindata.
- ② lcd → Recive character only (49)
- ③ lcd → Handle 1 char at one time →
- ④ You Can send Your data on
  - → 8 bit → 0 → 7
  - → 4 bit → 4 → 7
  - ① 8 bit one time | 4 bit twice

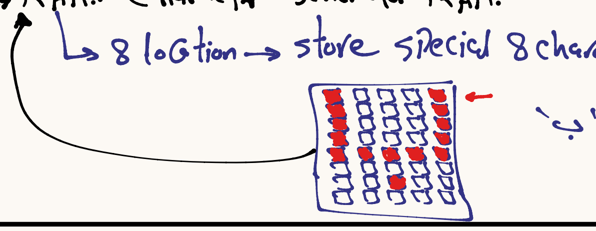
② CGRom: Character Generation Rom
- → lcD: 16*2
  - → No of digit
  - → location will display Character



0x01

* CGRom: store Display way for Character
  - → limition → Alph → Number → symbol
    - → Ascll table

③ CGRAM: Character Generator RAM.
- → 8 location → store special 8 character



DDRAM Address

line 1, Digit 10 (1, 10)

00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	
40	41	42	43	44	45	46	47	48	49	4A	4B	4C	4D	4E	4F	

Display Address | Hidden

0x00 + 10 → 0x0A

(line 2, 14) → 0x40 + 14 → 0x4D

CGRom
- → 0 → Ascll → ①
- → 48 → Ascll → 48 → 0

0011 0001

32 + 16 + 1 → 49

0x31

CGRAM → 8 location → 8 byte
- location → 0x00 (00–07)
- location → 0x08 (07–0F)

③ How to work (8 bit Mode)

↳ lcd → Display data
- → Rs → Data Display = 1
- → Rw → gnd ——→ 0
- → Update for your Data
  Pindata ← Data
- → E ——→ 1
- → wait ——→ 1Msec
- → E ——→ 0

00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
40	41	42	43	44	45	46	47	48	49	4A	4B	4C	4D	4E	4F

Move to (2, 4)

↳ $\boxed{0x40}$ + 4;

0x40 → 0b0100 0000

4 → 0b0000 0100 +

0100 0100

0x 4 4

LCD → Command

Rs ——→ Instruction ——→ 0
Rw ——→ gnd ——→ 0
Update Command.
Data Pin ——→ Command
E ——→ 1
wait 1Msec
E ——→ 0

LCD_Init (8 bit)

① wait 30Msec
② function set → Command → DL, NL, F
③ wait 1Msec
④ Display on-off → Command → D ✓, C ✓, B ✓
⑤ wait 1Msec
⑥ Clear
⑦ wait 2Msec
⑧ Entry Mode.

3 Bit → 8 option

D = 1	C = 0	B = 0
D = 0	C = 0	B = 0
D = 0	C = 1	B = 0
D = 0	C = 1	B = 1

Display off

4 ① Display off
② Display on ←
③ Cursor on
④ Cursor Blink

4.5 Display Commands

No.	Instructions	RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	Function
1	Clear Display	0	0	0	0	0	0	0	0	0	1	Write "20h" to DDRAM and set DDRAM address (AC) to "00h"
2	Return Home	0	0	0	0	0	0	0	0	1	x	Set DDRAM address (AC) to "00h" and return cursor to its original position if shifted (DDRAM contents are not change)
3	Entry Mode Set	0	0	0	0	0	0	0	1	I/D	S	Set cursor moving direction and specify display shift, during data read and write of DDRAM and CGRAM. S=1, screen shifting; S=0, no screen shifting I/D=1, AC=AC+1 and if S=1, screen shift left I/D=0, AC=AC-1 and if S=0, screen shift right
4	Display ON/OFF	0	0	0	0	0	0	1	D	C	B	D=1, display on; D=0, display off C=1, cursor on; C=0, cursor off B=1, cursor blinking on; B=0, cursor blinking off
5	Cursor or Display Shift	0	0	0	0	0	1	S/C	R/L	x	x	Move the cursor or shift the display, where DDRAM contents. S/C=1, shift screen; S/C=0, shift cursor R/L=1, to right-side; R/L=0, to left side (if S/C=1, AC will not be changed)
6	Function Set	0	0	0	0	1	DL	N	F	x	x	DL=1, 8-bit interface; DL=0, 4-bit interface N=1, 2-line display; N=0, 1-line display F=1, 5x11 dots font; F=0, 5x8 dots font
7	Set CGRAM address	0	0	0	1	AC5	AC4	AC3	AC2	AC1	AC0	Set CGRAM address in address counter
8	Set DDRAM address	0	0	1	AC6	AC5	AC4	AC3	AC2	AC1	AC0	Set DDRAM address in address counter
9	Read Busy flag & address	0	1	BF	AC6	AC5	AC4	AC3	AC2	AC1	AC0	Check the system status and get the address counter content (AC6~AC0). BF=1, busy; BF=0, ready
10	Write data to RAM	[illegible]	0	D7	D6	D5	D4	D3	D2	D1	D0	Write the data into internal RAM, where the address counter pointing at.
11	Read data from RAM	[illegible]	1	D7	D6	D5	D4	D3	D2	D1	D0	Read the data from internal RAM, where the address counter pointing at.

7 … 0

0x01

Test

Normal English

shift

Display
Cursor
Blink

Data length
No of line
Font size

LCD
8 bit
2 line
5X11
5X8

* Address → 0x08

CGRAM

01001000

Address

0x40 | 0x08

0100 1000

01000000
00001000
01001000

0x00
0x40

Address = 0x06 → DDRAM

1000 0110

0x80 | 0x06

1000 0000
0000 0110
1000 0110

* CGRAM → 0x40 | Add
DDRAM → 0x80 | Add